

LINEAR ALGEBRA

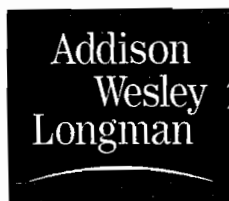
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Historical Notes by Victor J. Katz

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PREFACE

Our text is designed for use in a first undergraduate course in linear algebra. Because linear algebra provides the tools for dealing with problems in fields ranging from forestry to nuclear physics, it is desirable to make the subject accessible to students from a variety of disciplines. For the mathematics major, a course in linear algebra often serves as a bridge from the typical intuitive treatment of calculus to more rigorous courses such as abstract algebra and analysis. Recognizing this, we have attempted to achieve an appropriate blend of intuition and rigor in our presentation.

NEW FEATURES IN THIS EDITION

- **Evenly Paced Development:** In our previous edition, as in many other texts, fundamental notions of linear algebra—including linear combinations, subspaces, independence, bases, rank, and dimension—were first introduced in the context of axiomatically defined vector spaces. Having easily mastered matrix algebra and techniques for solving linear systems, many students were unable to adjust to the discontinuity in difficulty. In an attempt to eradicate this abrupt jump that has plagued instructors for years, we have extensively revised the first portion of our text to introduce these fundamental ideas gradually, in the context of $\mathbb{R}^n$. Thus linear combinations and spans of vectors are discussed in Section 1.1 and subspaces and bases are introduced where they first naturally occur in the study of the solution space of a homogeneous linear system. Independence, rank, dimension, and linear transformations are all introduced before axiomatic vector spaces are defined. The chapter dealing with vector spaces (Chapter 3) is only half its length in the previous edition, because the definitions, theorems, and proofs already given can usually be extended to general vector spaces by replacing “vectors in $\mathbb{R}^n$ ” by “vectors in a vector space V .” Most of the reorganization of our text was driven by our desire to tackle this problem.

- **Early Geometry:** Vector geometry (geometric addition, the dot product, length, the angle between vectors), which is familiar to many students from calculus, is now presented for vectors in $\mathbb{R}^n$ in the first two sections of Chapter 1. This provides a geometric foundation for notions of linear combinations and subspaces. Instructors may feel that students will be uncomfortable working immediately in $\mathbb{R}^n$. It is our experience that this causes no difficulty; students can compute a dot product of vectors with five components as easily as with two components.
- **Application to Coding:** An application of matrix algebra to binary linear codes now appears at the end of Chapter 1.
- **MATLAB:** The professional PC software MATLAB is widely used for computations in linear algebra. Throughout the text, we have included optional exercises to be done using the Student Edition of MATLAB. Each exercise set includes an explanation of the procedures and commands in MATLAB needed for the exercises. When performed sequentially throughout the text, these exercises give an elementary tutorial on MATLAB. Appendix D summarizes for easy reference procedures and commands used in the text exercises, but it is not necessary to study Appendix D before plunging in with the MATLAB exercises in Section 1.1. We have not written MATLAB .M-files combining MATLAB's commands for student use. Rather, we explain the MATLAB commands and ask students to type a single line, just as shown in the text, that combines the necessary commands, and then to edit and access that line as necessary for working a sequence of problems. We hope that students will grasp the commands, and proceed to write their own lines of commands to solve problems in this or other courses, referring if necessary to the MATLAB manual, which is the best reference. Once students have had some practice entering data, we do supply files containing matrix data for exercises to save time and avoid typos in data entry. For example, the data file for the MATLAB exercises for Section 1.4 is FBC1S4.M, standing for Fraleigh Beauregard Chapter 1 Section 4. The data files are on the disk containing LINTEK, as explained in the next item.
- **LINTEK:** The PC software LINTEK by Fraleigh, designed explicitly for this text, has been revised and upgraded and is free to students using our text. LINTEK is not designed for professional use. In particular, matrices can have no more than 10 rows or columns. LINTEK does provide some educational reinforcements, such as step-by-step execution and quizzes, that are not available in MATLAB. All information needed for LINTEK is supplied on screen. No manual is necessary. The matrix data files for MATLAB referred to in the preceding item also work with LINTEK, and are supplied on the LINTEK disk. Many optional exercise sets include problems to be done using LINTEK.

FEATURES RETAINED FROM THE PREVIOUS EDITION

- **Linear Transformations:** In the previous edition, we presented material on linear transformations throughout the text, rather than placing the topic toward the end of a one-semester course. This worked well. Our students gained much more understanding of linear transformations by working with them over a longer period of time. We have continued to do this in the present edition.
- **Eigenvalues and Eigenvectors:** This topic is introduced, with applications, in Chapter 5. Eigenvalues and eigenvectors recur in Chapters 7, 8, and 9, so students have the opportunity to continue to work with them.
- **Applications:** We believe that if applications are to be presented, it is best to give each one as soon as the requisite linear algebra has been developed, rather than deferring them all to the end of the text. Prompt work with an application reinforces the algebraic idea. Accordingly, we have placed applications at the ends of the chapters where the pertinent algebra is developed, unless the applications are so extensive that they merit a chapter by themselves. For example, Chapter 1 concludes with applications to population distribution (Markov chains) and to binary linear codes.
- **Summaries:** The summaries at the ends of sections have proved very convenient for both students and instructors, so we continue to provide them.
- **Exercises:** There are abundant pencil-and-paper exercises as well as computer exercises. Most exercise sets include a ten-part true-false problem. That exercise gives students valuable practice in deciding whether a mathematical statement is true, as opposed to asking for a proof of a given true statement. Answers to odd-numbered exercises having numerical answers are given at the back of the text. Usually, a requested proof or explanation is not given in the answers, because having it too readily available does not seem pedagogically sound. Computer related exercises are included at the end of most exercise sets. Their appearance is signaled by a disk logo.
- **Complex Numbers:** We use complex numbers when discussing eigenvalues and diagonalization in Chapter 5. Chapter 9 is devoted to linear algebra in $\mathbb{C}^n$. Some instructors bemoan a restriction to real numbers throughout most of the course, and they certainly have a point. We experimented when teaching from the previous edition, making the first section of the chapter on complex numbers the first lesson of the course. Then, as we developed real linear algebra, we always assigned an appropriate problem or two from the parallel development in subsequent sections on $\mathbb{C}^n$. Except for discussing the complex inner product and conjugate transpose, very little extra time was necessary. This technique proved to be feasible, but our students were not enamored with pencil-and-paper computations involving complex numbers.

- **Dependence Chart:** A dependence chart immediately follows this preface, and is a valuable aid in constructing a syllabus for the course.

SUPPLEMENTS

- **Instructor's Solutions Manual:** This manual, prepared by the authors, is available to the instructor from the publisher. It contains complete solutions, including proofs, for all of the exercises.
- **Student's Solutions Manual:** Prepared by the authors, this manual contains the complete solutions, including proofs, from the Instructor's Solutions Manual for every third problem (1, 4, 7, etc.) in each exercise set.
- **LINTEK:** This PC software, discussed above, is included with each copy of the text.
- **Testbank:** The authors have created a substantial test bank. It is available to the instructor from the publisher at no cost. Note that each multiple-choice problem in the bank can also be requested as open-ended—that is, with the choices omitted—as long as the problem still makes sense. Problems are easily selected and printed using software by Fraleigh. We use this bank extensively, saving ourselves much time—but of course, we made up the problems!

ACKNOWLEDGMENTS

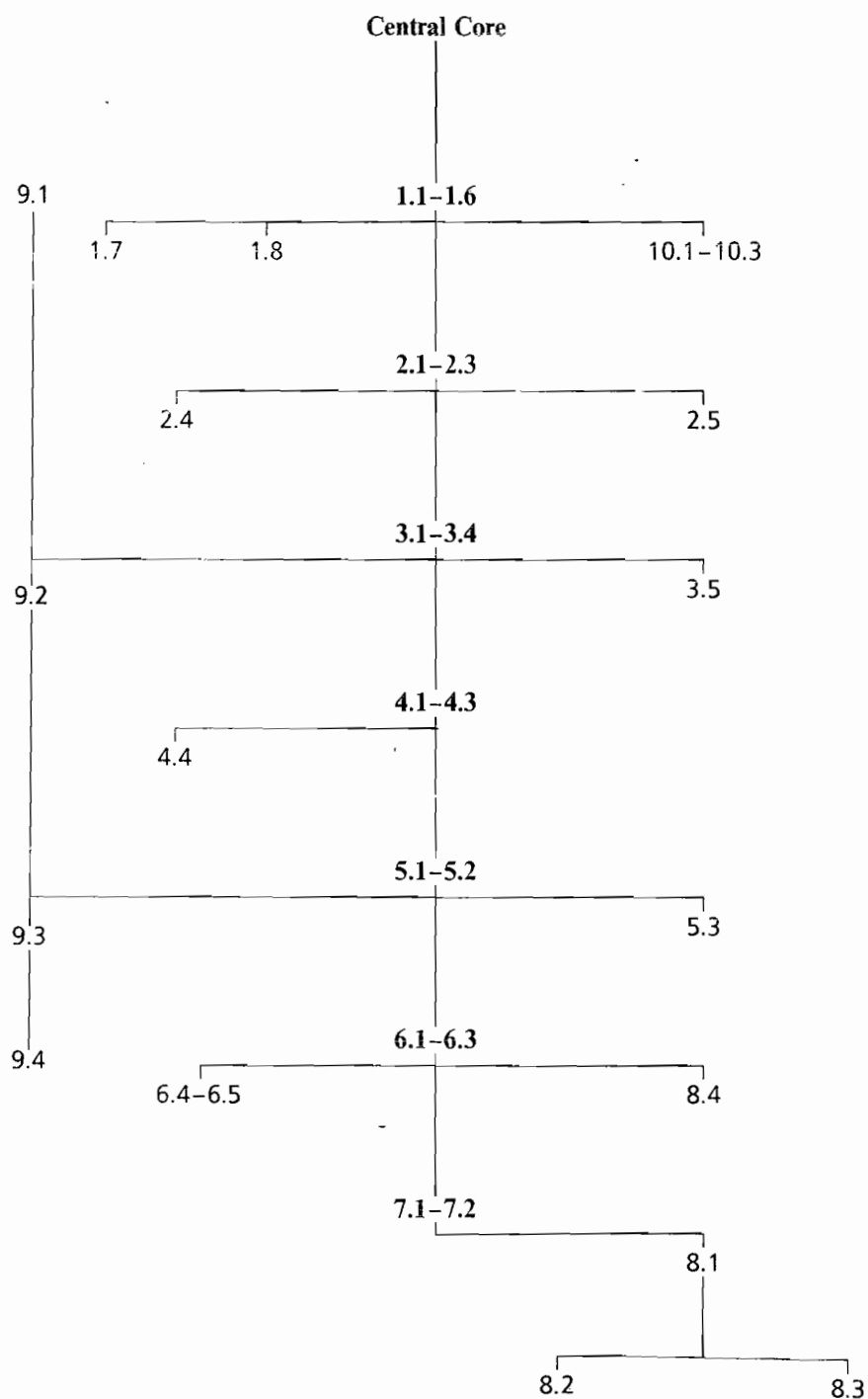
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DEPENDENCE CHART



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VECTORS, MATRICES, AND LINEAR SYSTEMS

We have all solved simultaneous linear equations—for example,

$$2x + y = 4$$

$$x - 2y = -3.$$

We shall call any such collection of simultaneous linear equations a *linear system*. Finding all solutions of a linear system is fundamental to the study of linear algebra. Indeed, the great practical importance of linear algebra stems from the fact that *linear systems can be solved by algebraic methods*. For example, a *linear* equation in one unknown, such as $3x = 8$, is easy to solve. But the nonlinear equations $x^5 + 3x = 1$, $x^x = 100$, and $x - \sin x = 1$ are all difficult to solve algebraically.

One often-used technique for dealing with a nonlinear problem consists of *linearizing the problem*—that is, approximating the problem with a linear one that can be solved more easily. Linearization techniques often involve calculus. If you have studied calculus, you may be familiar with Newton's method for approximating a solution to an equation of the form $f(x) = 0$; an example would be $x - 1 - \sin x = 0$. An approximate solution is found by solving sequentially several linear equations of the form $ax = b$, which are obtained by approximating the graph of f with lines. Finding an approximate numerical solution of a partial differential equation may involve solving a linear system consisting of thousands of equations in thousands of unknowns. With the advent of the computer, solving such systems is now possible. The feasibility of solving huge linear problems makes linear algebra currently one of the most useful mathematical tools in both the physical and the social sciences.

The study of linear systems and their solutions is phrased in terms of *vectors* and *matrices*. Sections 1.1 and 1.2 introduce vectors in the Euclidean spaces (the plane, 3-space, etc.) and provide a geometric foundation for our work. Sections 1.3–1.6 introduce matrices and methods for solving linear systems and study solution sets of linear systems.

1.1

VECTORS IN EUCLIDEAN SPACES

We all know the practicality of two basic arithmetic operations—namely, adding two numbers and multiplying one number by another. We can regard the real numbers as forming a line which is a one-dimensional space. In this section, we will describe a useful way of adding two points in a plane, which is a two-dimensional space, or two points in three-dimensional space. We will even describe what is meant by n -dimensional space and define addition of two points there. We will also describe how to multiply a point in two-, three-, and n -dimensional space by a real number. These extended notions of addition and of multiplication by a real number are as useful in n -dimensional space for $n > 1$ as they are for the one-dimensional real number line. When these operations are performed in spaces of dimension greater than one, it is conventional to call the elements of the space *vectors* as well as *points*. In this section, we describe a physical model that suggests the term *vector* and that motivates addition of vectors and multiplication of a vector by a number. We then formally define these operations and list their properties.

Euclidean Spaces

Let $\mathbb{R}$ be the set of all real numbers. We can regard $\mathbb{R}$ geometrically as the Euclidean line—that is, as **Euclidean 1-space**. We are familiar with rectangular x, y -coordinates in the Euclidean plane. We consider each ordered pair (a, b) of real numbers to represent a point in the plane, as illustrated in Figure 1.1. The set of all such ordered pairs of real numbers is **Euclidean 2-space**, which we denote by $\mathbb{R}^2$, and often call *the plane*.

To coordinatize space, we choose three mutually perpendicular lines as coordinate axes through a point that we call the *origin* and label 0 , as shown in Figure 1.2. Note that we represent only half of each coordinate axis for clarity. The coordinate system in this figure is called a *right-hand* system because, when the fingers of the right hand are curved in the direction required to rotate the positive x -axis toward the positive y -axis, the right thumb points up the z -axis, as shown in Figure 1.2. The set of all ordered triples (a, b, c) of real numbers is **Euclidean 3-space**, denoted $\mathbb{R}^3$, and often simply referred to as *space*.

Although a Euclidean space of dimension four or more may be difficult for us to visualize geometrically, we have no trouble writing down an ordered quadruple of real numbers such as $(2, -3, 7, \pi)$ or an ordered quintuple such as $(-0.3, 3, 2, -5, 21.3)$, etc. Indeed, it can be useful to do this. A household budget might contain nine categories, and the expenses allowed per week in each category could be represented by an ordered 9-tuple of real numbers. Generalizing, the set $\mathbb{R}^n$ of all ordered n -tuples $(x_1, x_2, \dots, x_n)$ of real numbers is **Euclidean n -space**. Note the use of just one letter with consecutive integer subscripts in this n -tuple, rather than different letters. We will often denote an element of $\mathbb{R}^2$ by (x_1, x_2) and an element of $\mathbb{R}^3$ by (x_1, x_2, x_3) .

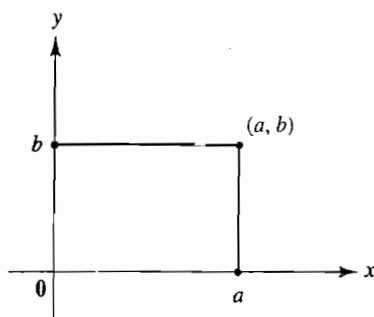


FIGURE 1.1
Rectangular coordinates in the plane.

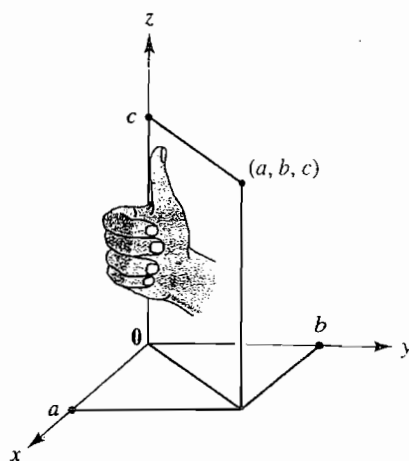


FIGURE 1.2
Rectangular coordinates in space.

The Physical Notion of a Vector

We are accustomed to visualizing an ordered pair or triple as a point in the plane or in space and denoting it geometrically by a dot, as shown in Figures 1.1 and 1.2. Physicists have found another very useful geometric interpretation of such pairs and triples in their consideration of forces acting on a body. The motion in response to a force depends on the *direction* in which the force is applied and on the *magnitude* of the force—that is, on how hard the force is exerted. It is natural to represent a force by an arrow, pointing in the direction

HISTORICAL NOTE THE IDEA OF AN n -DIMENSIONAL SPACE FOR $n > 3$ reached acceptance gradually during the nineteenth century; it is thus difficult to pinpoint a first “invention” of this concept. Among the various early uses of this notion are its appearances in a work on the divergence theorem by the Russian mathematician Mikhail Ostrogradskii (1801–1862) in 1836, in the geometrical tracts of Hermann Grassmann (1809–1877) in the early 1840s, and in a brief paper of Arthur Cayley (1821–1895) in 1846. Unfortunately, the first two authors were virtually ignored in their lifetimes. In particular, the work of Grassmann was quite philosophical and extremely difficult to read. Cayley’s note merely stated that one can generalize certain results to dimensions greater than three “without recourse to any metaphysical notion with regard to the possibility of a space of four dimensions.” Sir William Rowan Hamilton (1805–1865), in an 1841 letter, also noted that “it *must* be possible, in *some* way or other, to introduce not only triplets but *polyplets*, so as in some sense to satisfy the symbolical equation

$$a = (a_1, a_2, \dots, a_n);$$

a being here one symbol, as indicative of one (complex) thought; and $a_1, a_2, \dots, a_n$ denoting n real numbers, positive or negative.”

Hamilton, whose work on quaternions will be mentioned later, and who spent much of his professional life as the Royal Astronomer of Ireland, is most famous for his work in dynamics. As Erwin Schrödinger wrote, “the Hamiltonian principle has become the cornerstone of modern physics, the thing with which a physicist expects *every* physical phenomenon to be in conformity.”

in which the force is acting, and with the length of the arrow representing the magnitude of the force. Such an arrow is a *force vector*.

Using a rectangular coordinate system in the plane, note that if we consider a force vector to start from the origin $(0, 0)$, then the vector is completely determined by the coordinates of the point at the tip of the arrow. Thus we can consider each ordered pair in $\mathbb{R}^2$ to represent a vector in the plane as well as a point in the plane. When we wish to regard an ordered pair as a vector, we will use square brackets, rather than parentheses, to indicate this. Also, we often will write vectors as columns of numbers rather than as rows, and bracket notation is traditional for columns. Thus we speak of the *point* $(1, 2)$ in $\mathbb{R}^2$ and of the *vector* $[1, 2]$ in $\mathbb{R}^2$. To represent the *point* $(1, 2)$ in the plane, we make a dot at the appropriate place, whereas if we wish to represent the *vector* $[1, 2]$, we draw an arrow emanating from the origin with its tip at the place where we would plot the point $(1, 2)$. Mathematically, there is no distinction between $(1, 2)$ and $[1, 2]$. The different notations merely indicate different views of the same member of $\mathbb{R}^2$. This is illustrated in Figure 1.3. A similar observation holds for 3-space. Generalizing, each n -tuple of real numbers can be viewed both as a point $(x_1, x_2, \dots, x_n)$ and as a vector $[x_1, x_2, \dots, x_n]$ in $\mathbb{R}^n$. We use boldface letters such as $\mathbf{a} = [a_1, a_2]$, $\mathbf{v} = [v_1, v_2, v_3]$, and $\mathbf{x} = [x_1, x_2, \dots, x_n]$ to denote vectors. In written work, it is customary to place an arrow over a letter to denote a vector, as in $\vec{a}$, $\vec{v}$, and $\vec{x}$. The i th entry x_i in such a vector is the *i th component* of the vector. Even the real numbers in $\mathbb{R}$ can be regarded both as points and as vectors. When we are not regarding a real number as either a point or a vector, we refer to it as a *scalar*.

Two vectors $\mathbf{v} = [v_1, v_2, \dots, v_n]$ and $\mathbf{w} = [w_1, w_2, \dots, w_m]$ are **equal** if $n = m$ and $v_i = w_i$ for each i .

A vector containing only zeros as components is called a **zero vector** and is denoted by $\mathbf{0}$. Thus, in $\mathbb{R}^2$ we have $\mathbf{0} = [0, 0]$ whereas in $\mathbb{R}^4$ we have $\mathbf{0} = [0, 0, 0, 0]$.

When denoting a vector $\mathbf{v}$ in $\mathbb{R}^n$ geometrically by an arrow in a figure, we say that the vector is in *standard position* if it starts at the origin. If we draw an

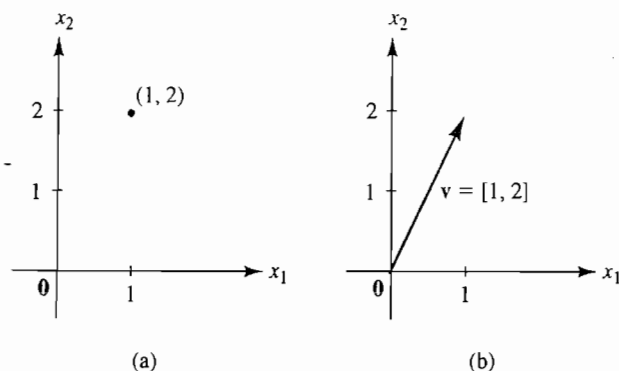


FIGURE 1.3

Two views of the same member of $\mathbb{R}^2$: (a) the point $(1, 2)$; (b) the vector $\mathbf{v} = [1, 2]$.

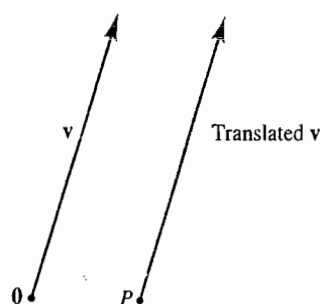


FIGURE 1.4
 v translated to P .

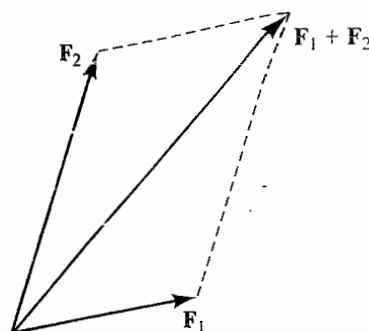


FIGURE 1.5
The vector sum $F_1 + F_2$.

arrow having the same length and parallel to the arrow representing v but starting at a point P other than the origin, we refer to the arrow as v translated to P . This is illustrated in Figure 1.4. Note that we did not draw any coordinate axes; we only marked the origin 0 and drew the two arrows. Thus we can consider Figure 1.4 to represent a vector v in $\mathbb{R}^2$, $\mathbb{R}^3$, or indeed in $\mathbb{R}^n$ for $n \geq 2$. We will often leave out axes when they are not necessary for our understanding. This makes our figures both less cluttered and more general.

Vector Algebra

Physicists tell us that if two forces corresponding to force vectors F_1 and F_2 act on a body at the same time, then the two forces can be replaced by a single force, the *resultant force*, which has the same effect as the original two forces. The force vector for this resultant force is the diagonal of the parallelogram having the force vectors F_1 and F_2 as edges, as illustrated in Figure 1.5. It is natural to consider this resultant force vector to be the *sum* $F_1 + F_2$ of the two original force vectors, and it is so labeled in Figure 1.5.

HISTORICAL NOTE THE CONCEPT OF A VECTOR in its earliest manifestation comes from physical considerations. In particular, there is evidence of velocity being thought of as a vector—a quantity with magnitude and direction—in Greek times. For example, in the treatise *Mechanica* by an unknown author in the fourth century B.C. is written: “When a body is moved in a certain ratio (i.e., has two linear movements in a constant ratio to one another), the body must move in a straight line, and this straight line is the diagonal of the parallelogram formed from the straight lines which have the given ratio.” Heron of Alexandria (first century A.D.) gave a proof of this result when the directions were perpendicular. He showed that if a point A moves with constant velocity over a line AB while at the same time the line AB moves with constant velocity along the parallel lines AC and BD so that it always remains parallel to its original position, and that if the time A takes to reach B is the same as the time AB takes to reach CD , then in fact the point A moves along the diagonal AD .

This basic idea of adding two motions vectorially was generalized from velocities to physical forces in the sixteenth and seventeenth centuries. One example of this practice is found as Corollary 1 to the Laws of Motion in Isaac Newton’s *Principia*, where he shows that “a body acted on by two forces simultaneously will describe the diagonal of a parallelogram in the same time as it would describe the sides by those forces separately.”

We can visualize two vectors with different directions and emanating from a point P in Euclidean 2-space or 3-space as determining a plane. It is pedagogically useful to do this for n -space for any $n \geq 2$ and show helpful figures on our pages. Motivated by our discussion of force vectors above, we consider the *sum* of two vectors $\mathbf{v}$ and $\mathbf{w}$ starting at a point P to be the vector starting at P that forms the diagonal of the parallelogram with a vertex at P and having edges represented by $\mathbf{v}$ and $\mathbf{w}$, as illustrated in Figure 1.6, where we take the vectors in $\mathbb{R}^n$ in standard position starting at $\mathbf{0}$. Thus we have a geometric understanding of *vector addition* in $\mathbb{R}^n$. We have labeled as *translated $\mathbf{v}$* and *translated $\mathbf{w}$* the sides of the parallelogram opposite the vectors $\mathbf{v}$ and $\mathbf{w}$.

Note that arrows along opposite sides of the parallelogram point in the same direction and have the same length. Thus, as a force vector, the translation of $\mathbf{v}$ is considered to be equivalent to the vector $\mathbf{v}$, and the same is true for $\mathbf{w}$ and its translation. We can think of obtaining the vector $\mathbf{v} + \mathbf{w}$ by drawing the arrow $\mathbf{v}$ from $\mathbf{0}$ and then drawing the arrow $\mathbf{w}$ translated to start from the tip of $\mathbf{v}$ as shown in Figure 1.6. The vector from $\mathbf{0}$ to the tip of the translated $\mathbf{w}$ is then $\mathbf{v} + \mathbf{w}$. This is often a useful way to regard $\mathbf{v} + \mathbf{w}$. To add three vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ geometrically, we translate $\mathbf{v}$ to start at the tip of $\mathbf{u}$ and then translate $\mathbf{w}$ to start at the tip of the translated $\mathbf{v}$. The sum $\mathbf{u} + \mathbf{v} + \mathbf{w}$ then begins at the origin where $\mathbf{u}$ starts, and ends at the tip of the translated $\mathbf{w}$, as indicated in Figure 1.7.

The difference $\mathbf{v} - \mathbf{w}$ of two vectors in $\mathbb{R}^n$ is represented geometrically by the arrow from the tip of $\mathbf{w}$ to the tip of $\mathbf{v}$, as shown in Figure 1.8. Here $\mathbf{v} - \mathbf{w}$ is the vector that, when added to $\mathbf{w}$, yields $\mathbf{v}$. The dashed arrow in Figure 1.8 shows $\mathbf{v} - \mathbf{w}$ in standard position.

If we are pushing a body with a force vector $\mathbf{F}$ and we wish to “double the force”—that is, we want to push in the same direction but twice as hard—then it is natural to denote the doubled force vector by $2\mathbf{F}$. If instead we want to push the body in the opposite direction with one-third the force, we denote the new force vector by $-\frac{1}{3}\mathbf{F}$. Generalizing, we consider the product $r\mathbf{v}$ of a scalar r times a vector $\mathbf{v}$ in $\mathbb{R}^n$ to be represented by the arrow whose length is $|r|$ times the length of $\mathbf{v}$ and which has the same direction as $\mathbf{v}$ if $r > 0$ but the opposite direction if $r < 0$. (See Figure 1.9 for an illustration.) Thus we have a geometric interpretation of *scalar multiplication* in $\mathbb{R}^n$ —that is, of multiplication of a vector in $\mathbb{R}^n$ by a scalar.

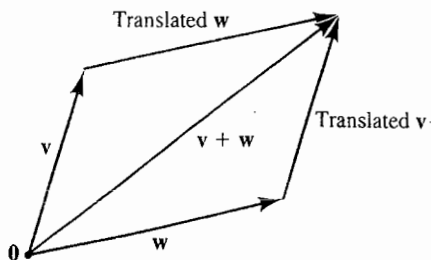


FIGURE 1.6
Representation of $\mathbf{v} + \mathbf{w}$ in $\mathbb{R}^n$.

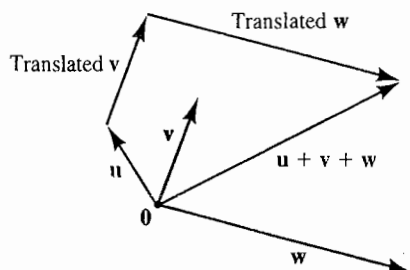


FIGURE 1.7
Representation of $\mathbf{u} + \mathbf{v} + \mathbf{w}$ in $\mathbb{R}^n$.

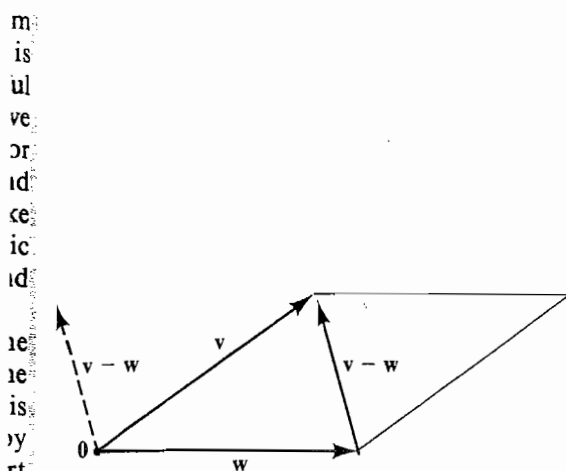


FIGURE 1.8
The vector $v - w$.

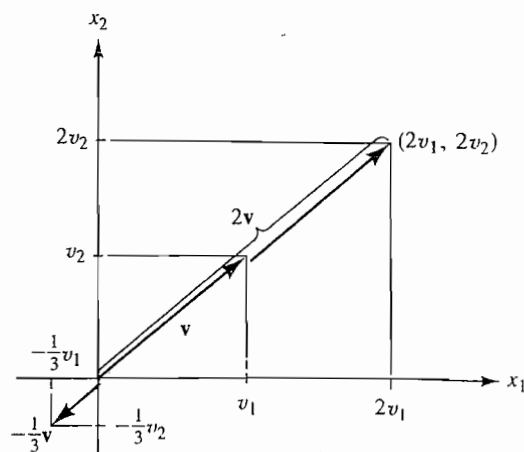


FIGURE 1.9
Computation of rv in $\mathbb{R}^2$.

Taking a vector $v = [v_1, v_2]$ in $\mathbb{R}^2$ and any scalar r , we would like to be able to compute rv algebraically as an element (ordered pair) in $\mathbb{R}^2$, and not just represent it geometrically by an arrow. Figure 1.9 shows the vector $2v$ which points in the same direction as v but is twice as long, and shows that we have $2v = [2v_1, 2v_2]$. It also indicates that if we multiply all components of v by $-\frac{1}{3}$, the resulting vector has direction opposite to the direction of v and length equal to $\frac{1}{3}$ the length of v . Similarly, if we take two vectors $v = [v_1, v_2]$ and $w = [w_1, w_2]$ in $\mathbb{R}^2$, we would like to be able to compute $v + w$ algebraically as an element (ordered pair) in $\mathbb{R}^2$. Figure 1.10 indicates that we have $v + w = [v_1 + w_1, v_2 + w_2]$ —that is, we can simply add corresponding components. With these figures to guide us, we formally define some algebraic operations with vectors in $\mathbb{R}^n$.

DEFINITION 1.1 Vector Algebra in $\mathbb{R}^n$

Let $v = [v_1, v_2, \dots, v_n]$ and $w = [w_1, w_2, \dots, w_n]$ be vectors in $\mathbb{R}^n$. The vectors are added and subtracted as follows:

Vector addition: $v + w = [v_1 + w_1, v_2 + w_2, \dots, v_n + w_n]$

Vector subtraction: $v - w = [v_1 - w_1, v_2 - w_2, \dots, v_n - w_n]$

If r is any scalar, the vector v is multiplied by r as follows:

Scalar multiplication: $rv = [rv_1, rv_2, \dots, rv_n]$

As a natural extension of Definition 1.1, we can combine three or more vectors in $\mathbb{R}^n$ using addition or subtraction by simply adding or subtracting their corresponding components. When a scalar in such a combination is negative, as in $4u + (-7)v + 2w$, we usually abbreviate by subtraction, writing $4u - 7v + 2w$. We write $-v$ for $(-1)v$.

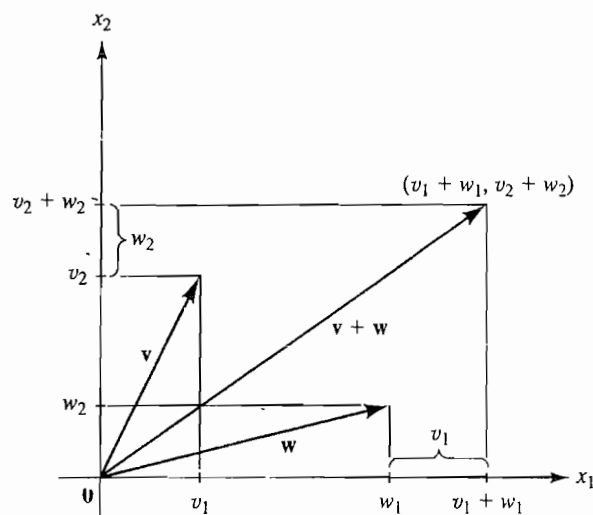


FIGURE 1.10
Computation of $v + w$ in $\mathbb{R}^2$.

EXAMPLE 1 Let $v = [-3, 5, -1]$ and $w = [4, 10, -7]$ in $\mathbb{R}^3$. Compute $5v - 3w$.

SOLUTION We compute

$$\begin{aligned} 5v - 3w &= 5[-3, 5, -1] - 3[4, 10, -7] \\ &= [-15, 25, -5] - [12, 30, -21] \\ &= [-27, -5, 16]. \end{aligned}$$

EXAMPLE 2 For vectors v and w in $\mathbb{R}^n$ pointing in different directions from the origin, represent geometrically $5v - 3w$.

SOLUTION This is done in Figure 1.11. ■

The analogues of many familiar algebraic laws for addition and multiplication of scalars also hold for vector addition and scalar multiplication. For convenience, we gather them in a theorem.

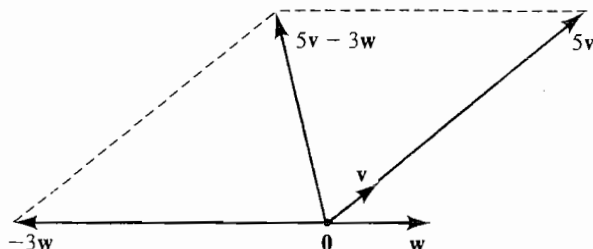


FIGURE 1.11
 $5v - 3w$ in $\mathbb{R}^n$.

THEOREM 1.1 Properties of Vector Algebra in $\mathbb{R}^n$

Let $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ be any vectors in $\mathbb{R}^n$, and let r and s be any scalars in $\mathbb{R}$.

Properties of Vector Addition

A1 $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$	An associative law
A2 $\mathbf{v} + \mathbf{w} = \mathbf{w} + \mathbf{v}$	A commutative law
A3 $\mathbf{0} + \mathbf{v} = \mathbf{v}$	$\mathbf{0}$ as additive identity
A4 $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$	$-\mathbf{v}$ as additive inverse of $\mathbf{v}$

Properties Involving Scalar Multiplication

S1 $r(\mathbf{v} + \mathbf{w}) = r\mathbf{v} + r\mathbf{w}$	A distributive law
S2 $(r + s)\mathbf{v} = r\mathbf{v} + s\mathbf{v}$	A distributive law
S3 $r(s\mathbf{v}) = (rs)\mathbf{v}$	An associative law
S4 $1\mathbf{v} = \mathbf{v}$	Preservation of scale

The eight properties given in Theorem 1.1 are quite easy to prove, and we leave most of them as exercises. The proofs in Examples 3 and 4 are typical.

EXAMPLE 3 Prove property A2 of Theorem 1.1.

SOLUTION Writing

$$\mathbf{v} = [v_1, v_2, \dots, v_n] \quad \text{and} \quad \mathbf{w} = [w_1, w_2, \dots, w_n],$$

we have

$$\mathbf{v} + \mathbf{w} = [v_1 + w_1, v_2 + w_2, \dots, v_n + w_n]$$

and

$$\mathbf{w} + \mathbf{v} = [w_1 + v_1, w_2 + v_2, \dots, w_n + v_n].$$

These two vectors are equal because $v_i + w_i = w_i + v_i$ for each i . Thus, the commutative law of vector addition follows directly from the commutative law of addition of numbers. ■

EXAMPLE 4 Prove property S2 of Theorem 1.1.

SOLUTION Writing $\mathbf{v} = [v_1, v_2, \dots, v_n]$, we have

$$\begin{aligned} (r + s)\mathbf{v} &= (r + s)[v_1, v_2, \dots, v_n] \\ &= [(r + s)v_1, (r + s)v_2, \dots, (r + s)v_n] \\ &= [rv_1 + sv_1, rv_2 + sv_2, \dots, rv_n + sv_n] \\ &= [rv_1, rv_2, \dots, rv_n] + [sv_1, sv_2, \dots, sv_n] \\ &= r\mathbf{v} + s\mathbf{v}. \end{aligned}$$

Thus the property $(r + s)\mathbf{v} = r\mathbf{v} + s\mathbf{v}$ involving vectors follows from the analogous property $(r + s)a_i = ra_i + sa_i$ for numbers. ■

Parallel Vectors

The geometric significance of multiplication of a vector by a scalar, as illustrated in Figure 1.9, leads us to this characterization of parallel vectors.

DEFINITION 1.2 Parallel Vectors

Two nonzero vectors $\mathbf{v}$ and $\mathbf{w}$ in $\mathbb{R}^n$ are **parallel**, and we write $\mathbf{v} \parallel \mathbf{w}$, if one is a scalar multiple of the other. If $\mathbf{v} = r\mathbf{w}$ with $r > 0$, then $\mathbf{v}$ and $\mathbf{w}$ have the **same direction**; if $r < 0$, then $\mathbf{v}$ and $\mathbf{w}$ have **opposite directions**.

EXAMPLE 5 Determine whether the vectors $\mathbf{v} = [2, 1, 3, -4]$ and $\mathbf{w} = [6, 3, 9, -12]$ are parallel.

SOLUTION We put $\mathbf{v} = r\mathbf{w}$ and try to solve for r . This gives rise to four component equations:

$$2 = 6r, \quad 1 = 3r, \quad 3 = 9r, \quad -4 = -12r.$$

Because $r = \frac{1}{3} > 0$ is a common solution to the four equations, we conclude that $\mathbf{v}$ and $\mathbf{w}$ are parallel and have the same direction. ■

Linear Combinations of Vectors

Definition 1.1 describes how to add or subtract two vectors, but as we remarked following the definition, we can use these operations to combine three or more vectors also. We give a formal extension of that definition.

DEFINITION 1.3 Linear Combination

Given vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ in $\mathbb{R}^n$ and scalars $r_1, r_2, \dots, r_k$ in $\mathbb{R}$, the vector

$$r_1\mathbf{v}_1 + r_2\mathbf{v}_2 + \cdots + r_k\mathbf{v}_k$$

is a **linear combination** of the vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ with **scalar coefficients** $r_1, r_2, \dots, r_k$.

The vectors $[1, 0]$ and $[0, 1]$ play a very important role in $\mathbb{R}^2$. Every vector $\mathbf{b}$ in $\mathbb{R}^2$ can be expressed as a linear combination of these two vectors in a *unique* way—namely, $\mathbf{b} = [b_1, b_2] = r_1[1, 0] + r_2[0, 1]$ if and only if $r_1 = b_1$ and $r_2 = b_2$. We call $[1, 0]$ and $[0, 1]$ the **standard basis vectors** in $\mathbb{R}^2$. They are often denoted by $\mathbf{i} = [1, 0]$ and $\mathbf{j} = [0, 1]$, as shown in Figure 1.12(a). Thus in $\mathbb{R}^2$, we may write the vector $[b_1, b_2]$ as $b_1\mathbf{i} + b_2\mathbf{j}$. Similarly, we have three **standard basis vectors** in $\mathbb{R}^3$ —namely,

$$\mathbf{i} = [1, 0, 0], \quad \mathbf{j} = [0, 1, 0], \quad \text{and} \quad \mathbf{k} = [0, 0, 1],$$

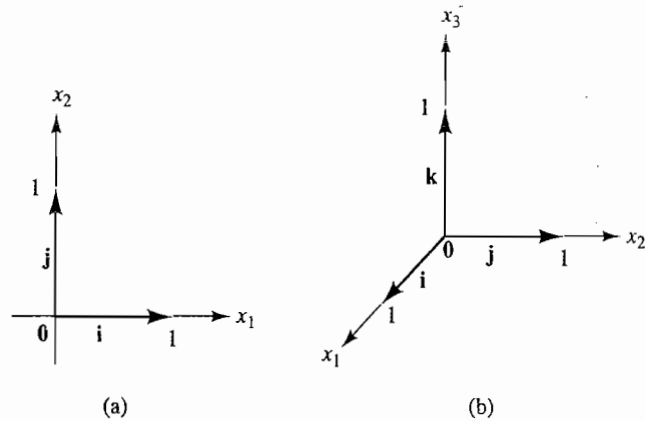


FIGURE 1.12

(a) Standard basis vectors in $\mathbb{R}^2$; (b) standard basis vectors in $\mathbb{R}^3$.

as shown in Figure 1.12(b). Every vector in $\mathbb{R}^3$ can be expressed uniquely as a linear combination of $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$. For example, we have $[3, -2, 6] = 3\mathbf{i} - 2\mathbf{j} + 6\mathbf{k}$. For $n > 3$, we denote the r th standard basis vector, having 1 as the r th component and zeros elsewhere, by

$$\mathbf{e}_r = [0, 0, \dots, 0, \underset{\substack{\uparrow \\ r\text{th component}}}{1}, 0, \dots, 0].$$

We then have

$$\mathbf{b} = [b_1, b_2, \dots, b_n] = b_1\mathbf{e}_1 + b_2\mathbf{e}_2 + \dots + b_n\mathbf{e}_n.$$

We see that every vector in $\mathbb{R}^n$ appears as a *unique* linear combination of the standard basis vector in $\mathbb{R}^n$.

The Span of Vectors

Let $\mathbf{v}$ be a vector in $\mathbb{R}^n$. All possible linear combinations of this single vector $\mathbf{v}$ are simply all possible scalar multiples $r\mathbf{v}$ for all scalars r . If $\mathbf{v} \neq \mathbf{0}$, all scalar multiples of $\mathbf{v}$ fill a line which we shall call the **line along $\mathbf{v}$** . Figure 1.13(a) shows the line along the vector $[-1, 2]$ in $\mathbb{R}^2$ while Figure 1.13(b) indicates the line along a nonzero vector $\mathbf{v}$ in $\mathbb{R}^n$.

Note that the line along $\mathbf{v}$ always contains the origin (the zero vector) because one scalar multiple of $\mathbf{v}$ is $0\mathbf{v} = \mathbf{0}$.

Now let $\mathbf{v}$ and $\mathbf{w}$ be two nonzero and nonparallel vectors in $\mathbb{R}^n$. All possible linear combinations of $\mathbf{v}$ and $\mathbf{w}$ are all vectors of the form $r\mathbf{v} + s\mathbf{w}$ for all scalars r and s . As indicated in Figure 1.14, all these linear combinations fill a plane which we call the **plane spanned by $\mathbf{v}$ and $\mathbf{w}$** .

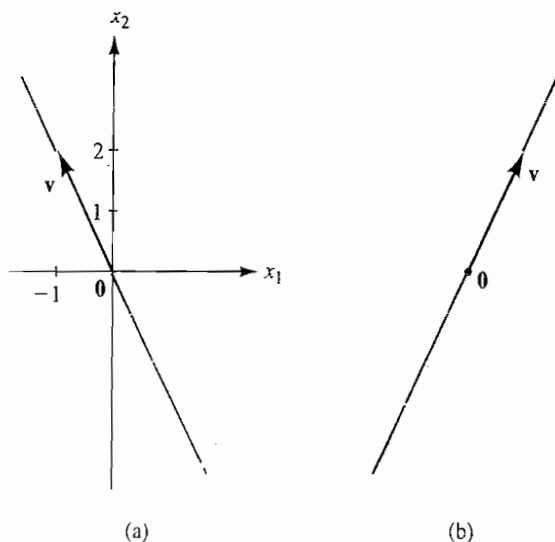


FIGURE 1.13

(a) The line along v in $\mathbb{R}^2$; (b) The line along v in $\mathbb{R}^n$.

EXAMPLE 6 Referring to Figure 1.15(a), estimate scalars r and s such that $rv + sw = b$ for the vectors v , w , and b all lying in the plane of the paper.

SOLUTION We draw the line along v , the line along w , and parallels to these lines through the tip of the vector b , as shown in Figure 1.15(b). From Figure 1.15(b), we estimate that $b = 1.5v - 2.5w$. ■

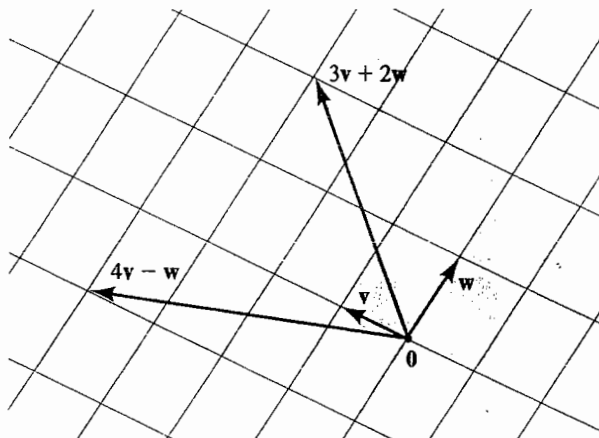


FIGURE 1.14

The plane spanned by v and w .

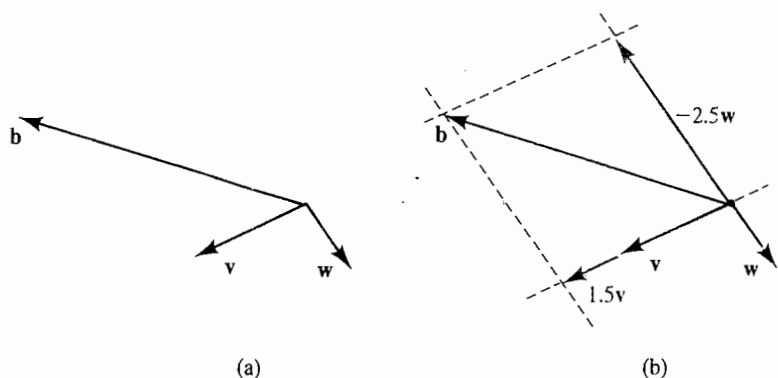


FIGURE 1.15

(a) Vectors v , w , and b ; (b) finding r and s so that $b = rv + sw$.

We now give an analytic analogue of Example 6 for two vectors in $\mathbb{R}^2$.

EXAMPLE 7 Let $v = [1, 3]$ and $w = [-2, 5]$ in $\mathbb{R}^2$. Find scalars r and s such that $rv + sw = [-1, 19]$.

SOLUTION Because $rv + sw = r[1, 3] + s[-2, 5] = [r - 2s, 3r + 5s]$, we see that $rv + sw = [-1, 19]$ if and only if both equations

$$r - 2s = -1$$

$$3r + 5s = 19$$

are satisfied. Multiplying the first equation by -3 and adding the result to the second equation, we obtain

$$0 + 11s = 22,$$

so $s = 2$. Substituting in the equation $r - 2s = -1$, we find that $r = 3$. ■

We note that the components -1 and 19 of the vector $[-1, 19]$ appear on the right-hand side of the system of two linear equations in Example 7. If we replace -1 by b_1 and 19 by b_2 , the same operations on the equations will enable us to solve for the scalars r and s in terms of b_1 and b_2 (see Exercise 42). This shows that all linear combinations of v and w do indeed fill the plane $\mathbb{R}^2$.

Example 7 indicates that an attempt to express a vector b as a linear combination of given vectors corresponds to an attempt to find a solution of a system of linear equations. This parallel is even more striking if we write our vectors as *columns* of numbers rather than as ordered *rows* of numbers—that is, as **column vectors** rather than as **row vectors**. For example, if we write the vectors v and w in Example 7 as columns so that

$$v = \begin{bmatrix} 1 \\ 3 \end{bmatrix} \quad \text{and} \quad w = \begin{bmatrix} -2 \\ 5 \end{bmatrix},$$

and also rewrite $[-1, 19]$ as a column vector, then the row-vector equation $r\mathbf{v} + s\mathbf{w} = [-1, 19]$ in the statement of Example 7 becomes

$$r \begin{bmatrix} 1 \\ 3 \end{bmatrix} + s \begin{bmatrix} -2 \\ 5 \end{bmatrix} = \begin{bmatrix} -1 \\ 19 \end{bmatrix}.$$

Notice that the numbers in this column-vector equation are in the same positions relative to each other as they are in the *system of linear equations*

$$\begin{aligned} r - 2s &= -1 \\ 3r + 5s &= 19 \end{aligned}$$

that we solved in Example 7. Every system of linear equations can be rewritten in this fashion as a single column-vector equation. Exercises 35–38 provide practice in this. Finding scalars $r_1, r_2, \dots, r_k$ such that $r_1\mathbf{v}_1 + r_2\mathbf{v}_2 + \dots + r_k\mathbf{v}_k = \mathbf{b}$ for given vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ and $\mathbf{b}$ in $\mathbb{R}^n$ is a fundamental computation in linear algebra. Section 1.4 describes an algorithm for finding all possible such scalars $r_1, r_2, \dots, r_k$.

The preceding paragraph indicates that often it will be natural for us to think of vectors in $\mathbb{R}^n$ as column vectors rather than as row vectors.

The **transpose** of a row vector $\mathbf{v}$ is defined to be the corresponding column vector, and is denoted by $\mathbf{v}^T$. Similarly, the transpose of a column vector is the corresponding row vector. For example,

$$[-1, 4, 15, -7]^T = \begin{bmatrix} -1 \\ 4 \\ 15 \\ -7 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 2 \\ -30 \\ 45 \end{bmatrix}^T = [2, -30, 45].$$

Note that for all vectors $\mathbf{v}$ we have $(\mathbf{v}^T)^T = \mathbf{v}$. As illustrated following Example 7, column vectors are often useful. In fact, some authors always regard every vector $\mathbf{v}$ in $\mathbb{R}^n$ as a column vector. Because it takes so much page space to write column vectors, these authors may describe $\mathbf{v}$ by giving the row vector $\mathbf{v}^T$. We do not follow this practice; we will write vectors in $\mathbb{R}^n$ as either row or column vectors depending on the context.

Continuing our geometric discussion, we expect that if $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are three nonzero vectors in $\mathbb{R}^n$ such that $\mathbf{u}$ and $\mathbf{v}$ are not parallel and also $\mathbf{w}$ is not a vector in the plane spanned by $\mathbf{u}$ and $\mathbf{v}$, then the set of all linear combinations of $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ will fill a three-dimensional portion of $\mathbb{R}^n$ —that is, a portion of $\mathbb{R}^n$ that looks just like $\mathbb{R}^3$. We consider the set of these linear combinations to be *spanned* by $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$. We make the following definition.

DEFINITION 1.4 Span of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$

Let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ be vectors in $\mathbb{R}^n$. The **span** of these vectors is the set of all linear combinations of them and is denoted by $\text{sp}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k)$. In set notation,

$$\text{sp}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k) = \{r_1\mathbf{v}_1 + r_2\mathbf{v}_2 + \dots + r_k\mathbf{v}_k \mid r_1, r_2, \dots, r_k \in \mathbb{R}\}.$$

It is important to note that $\text{sp}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k)$ in $\mathbb{R}^n$ may not fill what we intuitively consider to be a k -dimensional portion of $\mathbb{R}^n$. For example, in $\mathbb{R}^2$ we see that $\text{sp}([1, -2], [-3, 6])$ is just the one-dimensional line along $[1, -2]$ because $[-3, 6] = -3[1, -2]$ already lies in $\text{sp}([1, -2])$. Similarly, if $\mathbf{v}_3$ is a vector in $\text{sp}(\mathbf{v}_1, \mathbf{v}_2)$, then $\text{sp}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3) = \text{sp}(\mathbf{v}_1, \mathbf{v}_2)$ and so $\text{sp}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ is not three-dimensional. Section 2.1 will deal with this kind of *dependency* among vectors. As a result of our work there, we will be able to define dimensionality.

SUMMARY

1. *Euclidean n -space* $\mathbb{R}^n$ consists of all ordered n -tuples of real numbers. Each n -tuple $\mathbf{x}$ can be regarded as a *point* $(x_1, x_2, \dots, x_n)$ and represented graphically as a dot, or regarded as a vector $[x_1, x_2, \dots, x_n]$ and represented by an arrow. The n -tuple $\mathbf{0} = [0, 0, \dots, 0]$ is the *zero vector*. A real number $r \in \mathbb{R}$ is called a *scalar*.
2. Vectors $\mathbf{v}$ and $\mathbf{w}$ in $\mathbb{R}^n$ can be added and subtracted, and each can be multiplied by a scalar $r \in \mathbb{R}$. In each case, the operation is performed on the components, and the resulting vector is again in $\mathbb{R}^n$. Properties of these operations are summarized in Theorem 1.1. Graphic interpretations are shown in Figures 1.6, 1.8, and 1.9.
3. Two nonzero vectors in $\mathbb{R}^n$ are *parallel* if one is a scalar multiple of the other.
4. A *linear combination* of vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ in $\mathbb{R}^n$ is a vector of the form $r_1\mathbf{v}_1 + r_2\mathbf{v}_2 + \dots + r_k\mathbf{v}_k$, where each r_i is a scalar. The set of all such linear combinations is the *span* of the vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ and is denoted by $\text{sp}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k)$.
5. Every vector in $\mathbb{R}^n$ can be expressed uniquely as a linear combination of the *standard basis vectors* $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$, where $\mathbf{e}_i$ has 1 as its i th component and zeros for all other components.

EXERCISES

In Exercises 1–4, compute $\mathbf{v} + \mathbf{w}$ and $\mathbf{v} - \mathbf{w}$ for the given vectors $\mathbf{v}$ and $\mathbf{w}$. Then draw coordinate axes and sketch, using your answers, the vectors $\mathbf{v}$, $\mathbf{w}$, $\mathbf{v} + \mathbf{w}$, and $\mathbf{v} - \mathbf{w}$.

In Exercises 5–8, let $\mathbf{u} = [-1, 3, -2]$, $\mathbf{v} = [4, 0, -1]$, and $\mathbf{w} = [-3, -1, 2]$. Compute the indicated vector.

1. $\mathbf{v} = [2, -1]$, $\mathbf{w} = [-3, -2]$

2. $\mathbf{v} = [1, 3]$, $\mathbf{w} = [-2, 5]$

3. $\mathbf{v} = \mathbf{i} + 3\mathbf{j} + 2\mathbf{k}$, $\mathbf{w} = \mathbf{i} + 2\mathbf{j} + 4\mathbf{k}$

4. $\mathbf{v} = 2\mathbf{i} - \mathbf{j} + 3\mathbf{k}$, $\mathbf{w} = 3\mathbf{i} + 5\mathbf{j} + 4\mathbf{k}$

5. $3\mathbf{u} - 2\mathbf{v}$

6. $\mathbf{u} + 2(\mathbf{v} - 4\mathbf{w})$

7. $\mathbf{u} + \mathbf{v} - \mathbf{w}$

8. $4(3\mathbf{u} + 2\mathbf{v} - 5\mathbf{w})$

In Exercises 9–12, compute the given linear combination of $\mathbf{u} = [1, 2, 1, 0]$, $\mathbf{v} = [-2, 0, 1, 6]$, and $\mathbf{w} = [3, -5, 1, -2]$.

9. $\mathbf{u} - 2\mathbf{v} + 4\mathbf{w}$
10. $3\mathbf{u} + \mathbf{v} - \mathbf{w}$
11. $4\mathbf{u} - 2\mathbf{v} + 4\mathbf{w}$
12. $-\mathbf{u} + 5\mathbf{v} + 3\mathbf{w}$

In Exercises 13–16, reproduce on your paper those vectors in Figure 1.16 that appear in the exercise, and then draw an arrow representing each of the following linear combinations. All of the vectors are assumed to lie in the same plane. Use the technique illustrated in Figure 1.7 when all three vectors are involved.

13. $2\mathbf{u} + 3\mathbf{v}$
14. $-3\mathbf{u} + 2\mathbf{w}$
15. $\mathbf{u} + \mathbf{v} + \mathbf{w}$
16. $2\mathbf{u} - \mathbf{v} + \frac{1}{2}\mathbf{w}$

In Exercises 17–20, reproduce on your paper those vectors in Figure 1.17 that appear in the exercise, and then use the technique illustrated in Example 6 to estimate scalars r and s such that the given equation is true. All of the vectors are assumed to lie in the same plane.

17. $\mathbf{x} = r\mathbf{u} + s\mathbf{v}$
18. $\mathbf{y} = r\mathbf{u} + s\mathbf{v}$
19. $\mathbf{u} = r\mathbf{x} + s\mathbf{v}$
20. $\mathbf{y} = r\mathbf{u} + s\mathbf{x}$

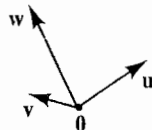


FIGURE 1.16

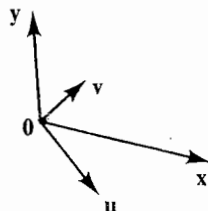


FIGURE 1.17

In Exercises 21–30, find all scalars c , if any exist, such that the given statement is true. Try to do some of these problems without using pencil and paper.

21. The vector $[2, 6]$ is parallel to the vector $[c, -3]$.
22. The vector $[c^2, -4]$ is parallel to the vector $[1, -2]$.
23. The vector $[c, -c, 4]$ is parallel to the vector $[-2, 2, 20]$.
24. The vector $[c^2, c^3, c^4]$ is parallel to the vector $[1, -2, 4]$ with the same direction.
25. The vector $[13, -15]$ is a linear combination of the vectors $[1, 5]$ and $[3, c]$.
26. The vector $[-1, c]$ is a linear combination of the vectors $[-3, 5]$ and $[6, -11]$.
27. $\mathbf{i} + c\mathbf{j} - 3\mathbf{k}$ is a linear combination of $\mathbf{i} + \mathbf{j}$ and $\mathbf{j} + 3\mathbf{k}$.
28. $\mathbf{i} + c\mathbf{j} + (c - 1)\mathbf{k}$ is in the span of $\mathbf{i} + 2\mathbf{j} + \mathbf{k}$ and $3\mathbf{i} + 6\mathbf{j} + 3\mathbf{k}$.
29. The vector $3\mathbf{i} - 2\mathbf{j} + c\mathbf{k}$ is in the span of $\mathbf{i} + 2\mathbf{j} - \mathbf{k}$ and $\mathbf{j} + 3\mathbf{k}$.
30. The vector $[c, -2c, c]$ is in the span of $[1, -1, 1]$, $[0, 1, -3]$, and $[0, 0, 1]$.

In Exercises 31–34, find the vector which, when translated, represents geometrically an arrow reaching from the first point to the second.

31. From $(-1, 3)$ to $(4, 2)$ in $\mathbb{R}^2$
32. From $(-3, 2, 5)$ to $(4, -2, -6)$ in $\mathbb{R}^3$
33. From $(2, 1, 5, -6)$ to $(3, -2, 1, 7)$ in $\mathbb{R}^4$

34. From $(1, 2, 3, 4, 5)$ to $(-5, -4, -3, -2, -1)$ in $\mathbb{R}^5$

35. Write the linear system

$$3x - 2y + 4z = 10$$

$$x - y - 3z = 0$$

$$2x + y - 5z = -3$$

as a column-vector equation.

36. Write the linear system

$$x_1 - 3x_2 + 2x_3 = -6$$

$$3x_1 - 4x_3 + 5x_4 = 12$$

as a column-vector equation.

37. Write the row-vector equation

$$p[-3, 4, 6] + q[0, -2, 5] - r[4, -3, 2] + s[6, 0, 7] = [8, -3, 1]$$

as

- a. a linear system,
b. a column-vector equation.

38. Write the column-vector equation

$$r_1 \begin{bmatrix} -2 \\ 3 \\ 0 \end{bmatrix} + r_2 \begin{bmatrix} 5 \\ 13 \\ -4 \end{bmatrix} + r_3 \begin{bmatrix} 16 \\ 0 \\ -9 \end{bmatrix} = \begin{bmatrix} 5 \\ -8 \\ 11 \end{bmatrix}$$

as a linear system.

39. Mark each of the following True or False.
- a. The notion of a vector in $\mathbb{R}^n$ is useful only if $n = 1, 2$, or 3 .
 - b. Every ordered n -tuple in $\mathbb{R}^n$ can be viewed both as a point and as a vector.
 - c. It would be impossible to define addition of points in $\mathbb{R}^n$ because we only add vectors.
 - d. If $\mathbf{a}$ and $\mathbf{b}$ are two vectors in standard position in $\mathbb{R}^n$, then the arrow from the tip of $\mathbf{a}$ to the tip of $\mathbf{b}$ is a translated representation of the vector $\mathbf{a} - \mathbf{b}$.

- e. If $\mathbf{a}$ and $\mathbf{b}$ are two vectors in standard position in $\mathbb{R}^n$, then the arrow from the tip of $\mathbf{a}$ to the tip of $\mathbf{b}$ is a translated representation of the vector $\mathbf{b} - \mathbf{a}$.
- f. The span of any two nonzero vectors in $\mathbb{R}^2$ is all of $\mathbb{R}^2$.
- g. The span of any two nonzero, nonparallel vectors in $\mathbb{R}^2$ is all of $\mathbb{R}^2$.
- h. The span of any three nonzero, nonparallel vectors in $\mathbb{R}^3$ is all of $\mathbb{R}^3$.
- i. If $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ are vectors in $\mathbb{R}^2$ such that $\text{sp}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k) = \mathbb{R}^2$, then $k = 2$.
- j. If $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ are vectors in $\mathbb{R}^3$ such that $\text{sp}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k) = \mathbb{R}^3$, then $k \geq 3$.

40. Prove the indicated property of vector addition in $\mathbb{R}^n$, stated in Theorem 1.1.

- a. Property A1
b. Property A3
c. Property A4

41. Prove the indicated property of scalar multiplication in $\mathbb{R}^n$, stated in Theorem 1.1.

42. a. Property S1
b. Property S3
c. Property S4

42. Prove algebraically that the linear system

$$r - 2s = b_1$$

$$3r + 5s = b_2$$

has a solution for all numbers $b_1, b_2 \in \mathbb{R}$, as asserted in the text.



43. Option 1 of the routine VECTGRPH in the software package LINTEK gives graphic quizzes on addition and subtraction of vectors in $\mathbb{R}^2$. Work with this Option 1 until you consistently achieve a score of 80% or better on the quizzes.

44. Repeat Exercise 43 using Option 2 of the routine VECTGRPH. The quizzes this time are on linear combinations in $\mathbb{R}^2$, and are quite similar to Exercises 17–20.

MATLAB

The MATLAB exercises are designed to build some familiarity with this widely used software as you work your way through the text. Complete information can be obtained from the manual that accompanies MATLAB. Some information is

available on the screen by typing **help** followed by a space and then the word or symbol about which you desire information.

The software LINTEK designed explicitly for this text does not require a manual, because all information is automatically given on the screen. However, MATLAB is a professionally designed program that is much more powerful than LINTEK. Although LINTEK adequately illustrates material in the text, the prospective scientist would be well advised to invest the extra time necessary to acquire some facility with MATLAB.

*Access MATLAB according to the directions for your installation. The MATLAB prompt, which is its request for instructions, looks like $\gg$. Vectors can be entered by entering the components in square brackets, separated by spaces; do not use commas. Enter vectors **a**, **b**, **c**, **u**, **v**, and **w** by typing the lines displayed below. After you type each line, press the Enter key. The vector components will be displayed, without brackets, using decimal notation. Proofread each vector after you enter it. If you have made an error, strike the up-arrow key and edit in the usual fashion to correct your error. (If you do not want a vector displayed for proofreading after entry, you can achieve this by typing `;` after the closing square bracket.) If you ever need to continue on the next line to type data, enter at least two periods `..` and immediately press the Enter key and continue the data.*

```
a = [2 -4 5 7]
b = [-1 6 7 3]
c = [13 -21 5 39]
u = [2/3 3/5 1/7]
v = [3/2 -5/6 11/3]
w = [5/7 3/4 -2/3]
```

*Now enter **u** (that is, type **u** and press the Enter key) to see **u** displayed again. Then enter `format long` and then **u** to see the components of **u** displayed with more decimal places. Enter `format short` and then **u** to return to the display with fewer decimal places. Enter `rat(u, 's')`, which displays rational approximations that are accurate when the numbers involved are fractions with sufficiently small denominators, to see **u** displayed again in fraction (rational) format. Addition and subtraction of vectors can be performed using `+` and `-`, and scalar multiplication using `*`. Enter `a + b` to see this sum displayed. Then enter `-3*a` to see this scalar product displayed. Using what you have discovered about MATLAB, work the following exercises. Entering `who` at any time displays a list of variables to which you have assigned numerical, vector, or matrix values. When you have finished, enter `quit` or `exit` to leave MATLAB.*

- M1. Compute $2\mathbf{a} - 3\mathbf{b} - 5\mathbf{c}$.
- M2. Compute $3\mathbf{c} - 4(2\mathbf{a} - \mathbf{b})$.
- M3. Attempt to compute $\mathbf{a} + \mathbf{u}$. What happens, and why?
- M4. Compute $\mathbf{u} + \mathbf{v}$ in
 - a. short format
 - b. long format
 - c. rational (fraction) format.
- M5. Repeat Exercise M4 for $2\mathbf{u} - 3\mathbf{v} + \mathbf{w}$ by first entering `x = 2*u - 3*v + w` and then looking at **x** in the different formats.

- M6. Repeat Exercise M4 for $\frac{1}{3}\mathbf{a} - \frac{3}{7}\mathbf{b}$, entering the fractions as (1/3) and (3/7), and using the time-saving technique of Exercise M5.
- M7. Repeat Exercise M4 for $0.3\mathbf{u} - 0.23\mathbf{w}$, using the time-saving technique of Exercise M5.
- M8. The *transpose* $\mathbf{v}^T$ of a vector $\mathbf{v}$ is denoted in MATLAB by $\mathbf{v}'$. Compute $\mathbf{a}^T - 3\mathbf{c}^T$ and $(\mathbf{a} - 3\mathbf{c})^T$. How do they compare?
- M9. Try to compute $\mathbf{u} + \mathbf{u}^T$. What happens, and why?
- M10. Enter **help** : to see some of the things that can be done using the colon. Use the colon in a statement starting $\mathbf{v1} =$ to generate the vector $\mathbf{v1} = [-3 \ -2 \ -1 \ 0 \ 1 \ 2]$, and then generate $\mathbf{v2} = [1 \ 4 \ 7 \ 10 \ 13 \ 16]$ similarly. Compute $\frac{1}{3}\mathbf{v1} + \frac{5}{6}\mathbf{v2}$ in rational format. (We use $\mathbf{v1}$ and $\mathbf{v2}$ in MATLAB where we would use v_1 and v_2 in the text.)
- M11. MATLAB can provide graphic illustrations of the component values of a vector. Enter **plot(a)** and note how the figure drawn reflects the components of the vector $\mathbf{a} = [2 \ -4 \ 5 \ 7]$. Press any key to clear the screen, and repeat the experiment with **bar(a)** and finally with **stairs(a)** to see two other ways to illustrate the component values of the vector $\mathbf{a}$.
- M12. Using the **plot** command, we can plot graphs of functions in MATLAB. The command **plot(x, y)** will plot the x vector against the y vector. Enter $\mathbf{x} = -1: .5: 1$. Note that, using the colons, we have generated a vector of x -coordinates starting at -1 and stopping at 1 , with increments of 0.5 . Now enter $\mathbf{y} = \mathbf{x} . * \mathbf{x}$. (A period before an operator, such as $+$ or $*$, will cause that operation to be performed on each component of a vector. Enter **help** . to see MATLAB explain this.) You will see that the y vector contains the squares of the x -coordinates in the x vector. Enter **plot(x, y)** to see a crude plot of the graph of $y = x^2$ for $-1 \leq x \leq 1$.
- Use the up-arrow key to return to the colon statement and make the increment $.2$ rather than $.5$ to get a better graph. Use the up-arrow key to get to the $\mathbf{y} = \mathbf{x} . * \mathbf{x}$ command, and press the Enter key to generate the new y vector with more entries. Then get to the **plot(x, y)** command and press Enter to see the improved graph of $y = x^2$ for $-1 \leq x \leq 1$.
 - Proceed as in part (a) to graph $y = x^2$ for $-3 \leq x \leq 3$ with increments of 0.2 . This time, put a semicolon after the command that defines the vector $\mathbf{x}$ before pressing the Enter key, so that you don't see the x -coordinates printed out. Similarly, put a semicolon after the command defining the vector $\mathbf{y}$.
 - Plot the graph of $y = \sin(x)$ for $-4\pi \leq x \leq 4\pi$. The number π can be entered as **pi** in MATLAB. Remember to use $*$ for multiplication.
 - Plot the graph of $y = 3 \cos(2x) - 2 \sin(x)$ for $-4\pi \leq x \leq 4\pi$. Remember to use $*$ for multiplication.

1.2

THE NORM AND THE DOT PRODUCT

The Magnitude of a Vector

The *magnitude* $\|\mathbf{v}\|$ of $\mathbf{v} = [v_1, v_2]$ is considered to be the length of the arrow in Figure 1.18. Using the Pythagorean theorem, we have

$$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2}. \quad (1)$$

EXAMPLE 1 Represent the vector $\mathbf{v} = [3, -4]$ geometrically, and find its magnitude.

SOLUTION The vector $[3, -4]$ has magnitude

$$\|\mathbf{v}\| = \sqrt{3^2 + (-4)^2} = \sqrt{25} = 5$$

and is shown in Figure 1.19. ■

In Figure 1.20, the *magnitude* $\|\mathbf{v}\|$ of a vector $\mathbf{v} = [v_1, v_2, v_3]$ in $\mathbb{R}^3$ appears as the length of the hypotenuse of a right triangle whose altitude is $|v_3|$ and whose base in the x_1, x_2 -plane has length $\sqrt{v_1^2 + v_2^2}$. Using the Pythagorean theorem, we obtain

$$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + v_3^2}. \quad (2)$$

EXAMPLE 2 Represent the vector $\mathbf{v} = [2, 3, 4]$ geometrically, and find its magnitude.

SOLUTION The vector $\mathbf{v} = [2, 3, 4]$ has magnitude $\|\mathbf{v}\| = \sqrt{2^2 + 3^2 + 4^2} = \sqrt{29}$ and is represented in Figure 1.21. ■

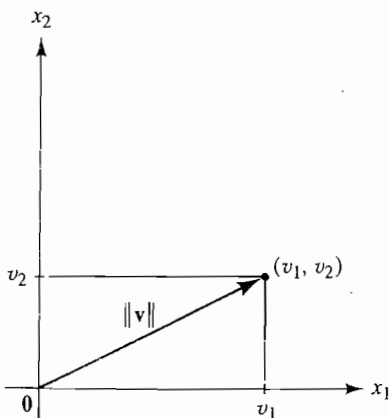


FIGURE 1.18
The magnitude of $\mathbf{v}$ in $\mathbb{R}^2$.

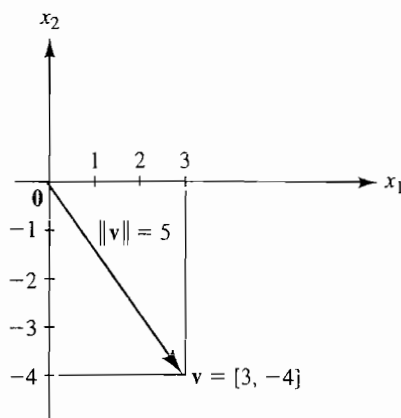


FIGURE 1.19
The magnitude of $[3, -4]$.

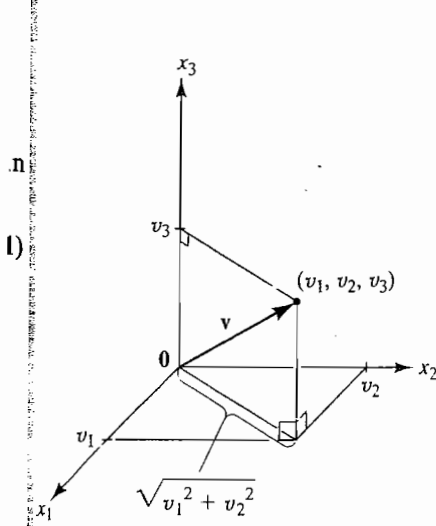


FIGURE 1.20
The magnitude of $\mathbf{v}$ in $\mathbb{R}^3$.

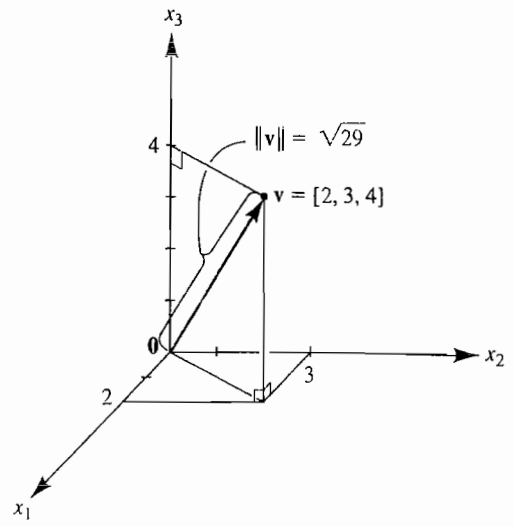


FIGURE 1.21
The magnitude of $[2, 3, 4]$.

The magnitude of a vector is also called the *norm* or the *length* of the vector. As suggested by Eqs. (1) and (2), we define the norm $\|\mathbf{v}\|$ of a vector $\mathbf{v}$ in $\mathbb{R}^n$ as follows.

DEFINITION 1.5 Norm or Magnitude of a Vector in $\mathbb{R}^n$

Let $\mathbf{v} = [v_1, v_2, \dots, v_n]$ be a vector in $\mathbb{R}^n$. The **norm** or **magnitude** of $\mathbf{v}$ is

$$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}. \quad (3)$$

EXAMPLE 3 Find the magnitude of the vector $\mathbf{v} = [-2, 1, 3, -1, 4, 2, 1]$.

SOLUTION We have

$$\|\mathbf{v}\| = \sqrt{(-2)^2 + 1^2 + 3^2 + (-1)^2 + 4^2 + 2^2 + 1^2} = \sqrt{36} = 6. \quad \blacksquare$$

Here are some properties of this norm operation.

THEOREM 1.2 Properties of the Norm in $\mathbb{R}^n$

For all vectors $\mathbf{v}$ and $\mathbf{w}$ in $\mathbb{R}^n$ and for all scalars r , we have

1. $\|\mathbf{v}\| \geq 0$ and $\|\mathbf{v}\| = 0$ if and only if $\mathbf{v} = \mathbf{0}$ **Positivity**
2. $\|r\mathbf{v}\| = |r| \|\mathbf{v}\|$ **Homogeneity**
3. $\|\mathbf{v} + \mathbf{w}\| \leq \|\mathbf{v}\| + \|\mathbf{w}\|$ **Triangle inequality**

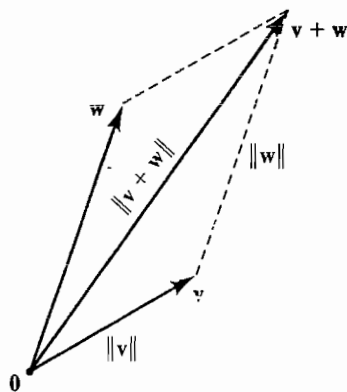


FIGURE 1.22
The triangle inequality.

Proofs of Properties 1 and 2 follow immediately from Definition 1.5 and appear as exercises at the end of this section. Figure 1.22 shows why Property 3 is called the triangle inequality; geometrically, it states that the length of a side of a triangle is less than or equal to the sum of the lengths of the other two sides. Although this seems obvious to us from Figure 1.22, we really should prove it—at least for $n > 3$, where we simply extended our definition of $\|v\|$ for v in $\mathbb{R}^2$ or $\mathbb{R}^3$ without any further geometric justification. A proof of the triangle inequality is given at the close of this section.

Unit Vectors

A vector in $\mathbb{R}^n$ is a **unit vector** if it has magnitude 1. Given any nonzero vector v in $\mathbb{R}^n$, a unit vector having the same direction as v is given by $(1/\|v\|)v$.

EXAMPLE 4 Find a unit vector having the same direction as $v = [2, 1, -3]$, and find a vector of magnitude 3 having direction opposite to v .

SOLUTION Because $\|v\| = \sqrt{2^2 + 1^2 + (-3)^2} = \sqrt{14}$, we see that $u = (1/\sqrt{14})[2, 1, -3]$ is the unit vector with the same direction as v , and $-3u = (-3/\sqrt{14})[2, 1, -3]$ is the other required vector. ■

The two-component unit vectors are precisely the vectors that extend from the origin to the unit circle $x^2 + y^2 = 1$ with center $(0, 0)$ and radius 1 in $\mathbb{R}^2$. (See Figure 1.23a.) The three-component unit vectors extend from $(0, 0, 0)$ to the unit sphere in $\mathbb{R}^3$, as illustrated in Figure 1.23(b).

Note that the standard basis vectors i and j in $\mathbb{R}^2$, as well as i, j , and k in $\mathbb{R}^3$, are unit vectors. In fact, the standard basis vectors $e_1, e_2, \dots, e_n$ for $\mathbb{R}^n$ are unit vectors, because each has zeros in all components except for one component of 1. For this reason, these standard basis vectors are also called **unit coordinate vectors**.

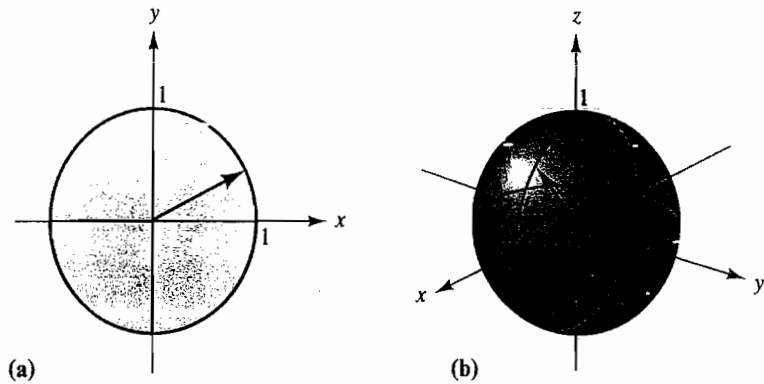


FIGURE 1.23

(a) Typical unit vector in $\mathbb{R}^2$; (b) typical unit vector in $\mathbb{R}^3$.

The Dot Product

The dot product of two vectors is a scalar that we will encounter as we now try to define the angle θ between two vectors $\mathbf{v} = [v_1, v_2, \dots, v_n]$ and $\mathbf{w} = [w_1, w_2, \dots, w_n]$ in $\mathbb{R}^n$, shown symbolically in Figure 1.24. To motivate the definition of θ , we will use the law of cosines for the triangle symbolized in Figure 1.24. Using our definition of the norm of a vector in $\mathbb{R}^n$ to compute the lengths of the sides of the triangle, the law of cosines yields

$$\|\mathbf{v}\|^2 + \|\mathbf{w}\|^2 = \|\mathbf{v} - \mathbf{w}\|^2 + 2\|\mathbf{v}\| \|\mathbf{w}\| (\cos \theta)$$

or

$$\begin{aligned} v_1^2 + \dots + v_n^2 + w_1^2 + \dots + w_n^2 \\ = (v_1 - w_1)^2 + \dots + (v_n - w_n)^2 + 2\|\mathbf{v}\| \|\mathbf{w}\| (\cos \theta). \end{aligned} \quad (4)$$

After computing the squares on the right-hand side of Eq. (4) and simplifying, we obtain

$$\|\mathbf{v}\| \|\mathbf{w}\| (\cos \theta) = v_1 w_1 + \dots + v_n w_n. \quad (5)$$

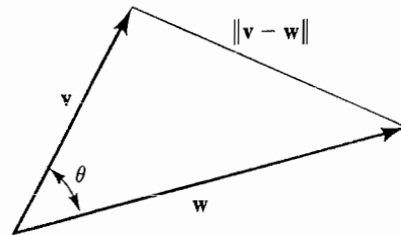


FIGURE 1.24

The angle between $\mathbf{v}$ and $\mathbf{w}$.

The sum of products of corresponding components in the vectors $\mathbf{v}$ and $\mathbf{w}$ on the right-hand side of Eq. (5) is frequently encountered, and is given a special name and notation.

DEFINITION 1.6 The Dot Product

The *dot product* of vectors $\mathbf{v} = [v_1, v_2, \dots, v_n]$ and $\mathbf{w} = [w_1, w_2, \dots, w_n]$ in $\mathbb{R}^n$ is the scalar given by

$$\mathbf{v} \cdot \mathbf{w} = v_1 w_1 + v_2 w_2 + \cdots + v_n w_n. \quad (6)$$

The dot product is sometimes called the *inner product* or the *scalar product*. To avoid possible confusion with scalar multiplication, we shall never use the latter term.

In view of Definition 1.6, we can write Eq. (5) as

$$\mathbf{v} \cdot \mathbf{w} = \|\mathbf{v}\| \|\mathbf{w}\| (\cos \theta). \quad (7)$$

Equation (7) suggests the following definition of the angle θ between two vectors $\mathbf{v}$ and $\mathbf{w}$ in $\mathbb{R}^n$.

The **angle** between nonzero vectors $\mathbf{v}$ and $\mathbf{w}$ is $\arccos \left(\frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\| \|\mathbf{w}\|} \right)$. (8)

Expression (8) makes sense, provided that

$$-1 \leq \frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\| \|\mathbf{w}\|} \leq 1, \quad (9)$$

so that we can indeed compute the arccosine of $(\mathbf{v} \cdot \mathbf{w})/(\|\mathbf{v}\| \|\mathbf{w}\|)$. This inequality (9) is usually rewritten in the form

$$|\mathbf{v} \cdot \mathbf{w}| \leq \|\mathbf{v}\| \|\mathbf{w}\|. \quad \text{Schwarz inequality} \quad (10)$$

We obtained it by assuming that Figure 1.24 is an appropriate representation for vectors $\mathbf{v}$ and $\mathbf{w}$ in $\mathbb{R}^n$. We give a purely algebraic proof of it at the end of this section to validate the definition in expression (8).

EXAMPLE 5 Find the angle θ between the vectors $[1, 2, 0, 2]$ and $[-3, 1, 1, 5]$ in $\mathbb{R}^4$.

SOLUTION We have

$$\cos \theta = \frac{[1, 2, 0, 2] \cdot [-3, 1, 1, 5]}{\sqrt{1^2 + 2^2 + 0^2 + 2^2} \sqrt{(-3)^2 + 1^2 + 1^2 + 5^2}} = \frac{9}{(3)(6)} = \frac{1}{2}.$$

Thus, $\theta = 60^\circ$. ■

Equation 7 gives a geometric meaning for the dot product.

The dot product of two vectors is equal to the product of their magnitudes with the cosine of the angle between them.

THEOREM 1.3 Properties of the Dot Product in $\mathbb{R}^n$

Let $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ be vectors in $\mathbb{R}^n$ and let r be any scalar in $\mathbb{R}$. The following properties hold:

- | | |
|--|------------------|
| D1 $\mathbf{v} \cdot \mathbf{w} = \mathbf{w} \cdot \mathbf{v}$, | Commutative law |
| D2 $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$, | Distributive law |
| D3 $r(\mathbf{v} \cdot \mathbf{w}) = (r\mathbf{v}) \cdot \mathbf{w} = \mathbf{v} \cdot (r\mathbf{w})$, | Homogeneity |
| D4 $\mathbf{v} \cdot \mathbf{v} \geq 0$, and $\mathbf{v} \cdot \mathbf{v} = 0$ if and only if $\mathbf{v} = \mathbf{0}$. | Positivity |

Verification of all of the properties in Theorem 1.3 is straightforward, as illustrated in the following example.

HISTORICAL NOTE THE SCHWARZ INEQUALITY is due independently to Augustin-Louis Cauchy (1789–1857) (see note on page 3), Hermann Amandus Schwarz (1843–1921), and Viktor Yakovlevich Bunyakovsky (1804–1889).

It was first stated as a theorem about coordinates in an appendix to Cauchy's 1821 text for his course on analysis at the Ecole Polytechnique, as follows:

$$|a\alpha + a'\alpha' + a''\alpha'' + \cdots| \leq \sqrt{a^2 + a'^2 + a''^2 + \cdots} \sqrt{\alpha^2 + \alpha'^2 + \alpha''^2 + \cdots}.$$

Cauchy's proof follows from the algebraic identity

$$\begin{aligned} (a\alpha + a'\alpha' + a''\alpha'' + \cdots)^2 &+ (a\alpha' - a'\alpha)^2 + (a\alpha'' - a''\alpha)^2 + \cdots + (a'\alpha'' - a''\alpha')^2 + \cdots \\ &= (a^2 + a'^2 + a''^2 + \cdots)(\alpha^2 + \alpha'^2 + \alpha''^2 + \cdots). \end{aligned}$$

Bunyakovsky proved the inequality for functions in 1859; that is, he stated the result

$$\left[\int_a^b f(x)g(x) dx \right]^2 \leq \int_a^b f^2(x) dx \cdot \int_a^b g^2(x) dx,$$

where we can consider $\int_a^b f(x)g(x) dx$ to be the inner product of the functions $f(x)$, $g(x)$ in the vector space of continuous functions on $[a, b]$. Bunyakovsky served as vice-president of the St. Petersburg Academy of Sciences from 1864 until his death. In 1875, the Academy established a mathematics prize in his name in recognition of his 50 years of teaching and research.

Schwarz stated the inequality in 1884. In his case, the vectors were functions ϕ , χ of two variables in a region T in the plane, and the inner product of these functions was given by $\iint_T \phi\chi dx dy$, where this integral is assumed to exist. The inequality then states that

$$\left| \iint_T \phi\chi dx dy \right| \leq \sqrt{\iint_T \phi^2 dx dy} \cdot \sqrt{\iint_T \chi^2 dx dy}.$$

Schwarz's proof is similar to the one given in the text (page 29). Schwarz was the leading mathematician in Berlin around the turn of the century; the work in which the inequality appears is devoted to a question about minimal surfaces.

EXAMPLE 6 Verify the positivity property D4 of Theorem 1.3.

SOLUTION We let $\mathbf{v} = [v_1, v_2, \dots, v_n]$, and we find that

$$\mathbf{v} \cdot \mathbf{v} = v_1^2 + v_2^2 + \cdots + v_n^2.$$

A sum of squares is nonnegative and can be zero if and only if each summand is zero. But a summand v_i^2 is itself a square, and will be zero if and only if $v_i = 0$. This completes the demonstration. ■

It is important to observe that the norm of a vector can be expressed in terms of its dot product with itself. Namely, for a vector $\mathbf{v}$ in $\mathbb{R}^n$ we have

$$\|\mathbf{v}\|^2 = \mathbf{v} \cdot \mathbf{v}. \quad (11)$$

Letting $\mathbf{v} = [v_1, v_2, \dots, v_n]$, we have

$$\mathbf{v} \cdot \mathbf{v} = v_1v_1 + v_2v_2 + \cdots + v_nv_n = \|\mathbf{v}\|^2.$$

Equation 11 enables us to use the algebraic properties of the dot product in Theorem 1.3 to prove things about the norm. This technique is illustrated in the proof of the Schwarz and triangle inequalities at the end of this section. Here is another illustration.

EXAMPLE 7 Show that the sum of the squares of the lengths of the diagonals of a parallelogram in $\mathbb{R}^n$ is equal to the sum of the squares of the lengths of the sides. (This is the *parallelogram relation*).

SOLUTION We take our parallelogram with vertex at the origin and with vectors $\mathbf{v}$ and $\mathbf{w}$ emanating from the origin to form two sides, as shown in Figure 1.25. The lengths of the diagonals are then $\|\mathbf{v} + \mathbf{w}\|$ and $\|\mathbf{v} - \mathbf{w}\|$. Using Eq. (11) and properties of the dot product, we have

$$\begin{aligned} \|\mathbf{v} + \mathbf{w}\|^2 + \|\mathbf{v} - \mathbf{w}\|^2 &= (\mathbf{v} + \mathbf{w}) \cdot (\mathbf{v} + \mathbf{w}) + (\mathbf{v} - \mathbf{w}) \cdot (\mathbf{v} - \mathbf{w}) \\ &= (\mathbf{v} \cdot \mathbf{v}) + 2(\mathbf{v} \cdot \mathbf{w}) + (\mathbf{w} \cdot \mathbf{w}) + (\mathbf{v} \cdot \mathbf{v}) - 2(\mathbf{v} \cdot \mathbf{w}) + (\mathbf{w} \cdot \mathbf{w}) \\ &= 2(\mathbf{v} \cdot \mathbf{v}) + 2(\mathbf{w} \cdot \mathbf{w}) \\ &= 2\|\mathbf{v}\|^2 + 2\|\mathbf{w}\|^2, \end{aligned}$$

which is what we wished to prove. ■

The definition of the angle θ between two vectors $\mathbf{v}$ and $\mathbf{w}$ in $\mathbb{R}^n$ leads naturally to this definition of perpendicular vectors, or *orthogonal* vectors as they are usually called in linear algebra.

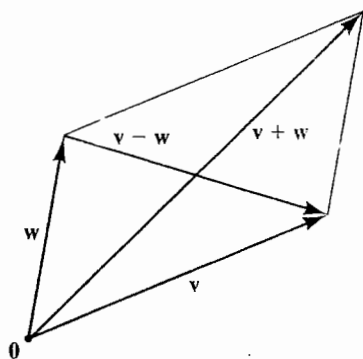


FIGURE 1.25

The parallelogram has $v + w$ and $v - w$ as vector diagonals.

DEFINITION 1.7 Perpendicular or Orthogonal Vectors

Two vectors v and w in $\mathbb{R}^n$ are **perpendicular** or **orthogonal**, and we write $v \perp w$, if $v \cdot w = 0$.

EXAMPLE 8 Determine whether the vectors $v = [4, 1, -2, 1]$ and $w = [3, -4, 2, -4]$ are perpendicular.

SOLUTION We have

$$v \cdot w = (4)(3) + (1)(-4) + (-2)(2) + (1)(-4) = 0.$$

Thus, $v \perp w$. ■

Application to Velocity Vectors and Navigation

The next two examples are concerned with another important physical vector model. A vector is the **velocity vector** of a moving object at an instant if it points in the direction of the motion and if its magnitude is the speed of the object at that instant. Physicists tell us that if a boat cruising with a heading and speed that would give it a still-water velocity vector s is also subject to a current that has velocity vector c , then the actual velocity vector of the boat is $v = s + c$.

EXAMPLE 9 Suppose that a ketch is sailing at 8 knots, following a course of 010° (that is, 10° east of north), on a bay that has a 2-knot current setting in the direction 070° (that is, 70° east of north). Find the course and speed made good. (The expression *made good* is standard navigation terminology for the actual course and speed of a vessel over the bottom.)

SOLUTION The velocity vectors s for the ketch and c for the current are shown in Figure 1.26, in which the vertical axis points due north. We find s and c by using a calculator and computing

$$s = [8 \cos 80^\circ, 8 \sin 80^\circ] \approx [1.39, 7.88]$$

and

$$c = [2 \cos 20^\circ, 2 \sin 20^\circ] \approx [1.88, 0.684].$$

By adding s and c , we find the vector v representing the course and speed of the ketch over the bottom—that is, the course and speed made good. Thus we have $v = s + c \approx [3.27, 8.56]$. Therefore, the speed of the ketch is

$$\|v\| \approx \sqrt{(3.27)^2 + (8.56)^2} \approx 9.16 \text{ knots,}$$

and the course made good is given by

$$90^\circ - \arctan\left(\frac{8.56}{3.27}\right) \approx 90^\circ - 69^\circ = 21^\circ.$$

That is, the course is 021° . ■

EXAMPLE 10 Suppose the captain of our ketch realizes the importance of keeping track of the current. He wishes to sail in 5 hours to a harbor that bears 120° and is 35 nautical miles away. That is, he wishes to make good the course 120° and the speed 7 knots. He knows from a tide and current table that the current is setting due south at 3 knots. What should be his course and speed through the water?

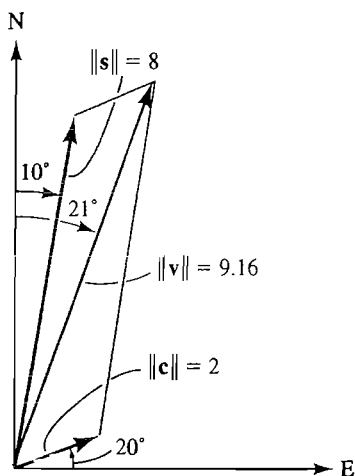


FIGURE 1.26
The vector $v = s + c$.

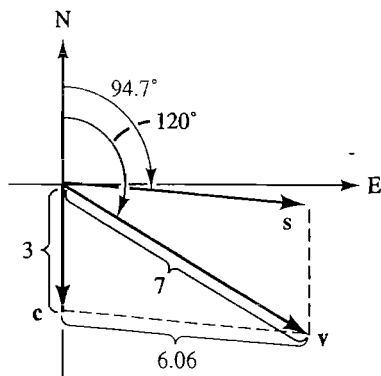


FIGURE 1.27
The vector $s = v - c$.